Green Ghar Score: M2 Mid-submission

Team Spades · Megathon 2026 · Status at 3:00 AM, 11 October · all figures below come from files in the repository (reports/, tests/)

1 Current approach and changes since M1

The approach is unchanged in shape: **score = fixed-weight sum of seven explainable sub-scores** (energy efficiency 25%, consumption intensity 20%, envelope 15%, cooling system 15%, windows and shading 10%, renewables 10%, smart use 5%), bands 80/65/50/30. Estimates come from a **physics model × LightGBM residual**, retrofits are applied **sequentially** so savings never double count, a **LambdaRank ranker** orders them, and a **hedonic value model with conformal intervals** prices the uplift. An LLM only narrates numbers the engine returns, and a verifier checks every figure. What M1 described as a plan is now built and running end to end:

Area	At M1 (plan)	Now (built)
Data	Synthetic homes only	50,000 synthetic homes (seed 42, 7 linked tables) **plus real public data in Supabase**: 80,811 rows from 9 sources (Census 2011 districts, NFHS-5 states, Open-Meteo weather, World Bank). Calibration views expose gaps, e.g. synthetic homes cook on clean fuel 100% of the time vs 71–99% in real urban NFHS-5 data.
Backend	FastAPI engine, local files	41 API paths. Supabase project (Postgres 17, 19 migrations, RLS, private bills bucket); the engine also exists as a SQL port with parity tests. Supabase JWT auth, owner-only access, rate limits, request logging.
AI Saathi	LLM with tools	Router → engine tools → **NVIDIA Nemotron 3.5 Lightning** (thinking on, streamed) → Groq → template fallback. New project-specific prompt, three few-shot examples and a 1,500-example SFT set, all built from the training split of our data. Hindi, Hinglish and English.
Frontend	Spec only	React 18 + TypeScript app served by the API at /app: Dashboard, Bill to score, Retrofit studio, Paisa (value and finance), AI Saathi, Garmi (heatwave) mode and a lender portfolio. Username and password sign-in and sign-up.
Channels	Bill OCR, Telegram	Bill reading for photos, text PDFs and scanned PDFs (Gemini, Tesseract fallback), masked consumer number, Telegram bot with signed deep links and polling mode.

Decisions changed since M1: (1) we calibrate against real public data instead of trusting the generator; (2) the LLM provider moved from Llama 3.3 to Nemotron, and because hosted models cannot be fine-tuned through the API we adapt it with prompt, few-shot and an SFT dataset instead; (3) login moved from email code to username and password.

2 Experiments run so far and their results

Split: by home, 70/15/15, seed 42. Model selection on validation only; numbers below are the held-out **test** set.

Experiment	Compared	Result (test)
E1 Annual electricity (4,041 homes)	Zone-median baseline, physics only, LightGBM only, physics × LightGBM	MAE 944 → 502 → 390 → **384 kWh**; MAPE 26.0% → **9.7%**; R^2 0.935. **59% lower error than baseline.**
E2 Retrofit ranking (4,000 homes, 37,170 pairs)	Popularity, rules (physics ROI), LambdaRank	NDCG@5 0.881 → 0.976 → **0.986**; Precision@3 0.720 → 0.904 → **0.942**. Infeasible items recommended: **0**.
E3 Savings accuracy	Physics only vs hybrid scaled to metered use	Electricity MAE 36.1 → **23.1 kWh**; direction of change correct in **100%** of cases; induction cooking correctly shown as raising electricity use.
E4 Home value uplift (12,470 sales)	City median ₹/sq ft, label rule, hedonic LightGBM	MAPE 11.5% / 29.6% → **6.7%**. 80% conformal interval covers **81.6%** at 24.2% width. Implied premium 1.63% per 10 points vs 1.5% assumed in the generator (self-consistency only, not real-world accuracy).
E5 Advisor grounding (180 held-out prompts, 6 categories)	Template mode; adversarial fake LLM that injects wrong numbers	Template: groundedness **100%**, key-fact accuracy **98.9%**. Adversarial: **139 injected hallucinations, 0 reached the user**; the verifier forced a safe fallback instead.

E6 Fairness check	Mean score by zone, dwelling type, tenure	Zones sit within 43.9–46.0 mean score; apartments score lower (40.2) because of roof access, which the advisor explains rather than hides.

Not yet measured: the live Nemotron groundedness and latency (our build sandbox blocks NVIDIA; the script `llm_eval.py --llm` is ready), and any accuracy on real customer bills (real data is used for calibration, not for training labels).

3 Progress on testing

Layer	Result	What it covers
Backend (pytest)	**95 passed**	No train/test leakage across homes; **monotonicity** (better equipment never lowers the score); fixed weights and tariff slabs; hidden scenarios (flats, tenants, missing data); SQL engine parity with Python; auth, ownership and rate limits; bill OCR including text and scanned PDFs; sign-up rules; verifier and advisor; Nemotron request and stream handling with mocked responses.
Frontend units (Vitest)	**23 passed**	API client (errors, retries, token refresh), figure formatting, login and sign-up validation, Garmi mode with real-weather payloads.
End to end (Playwright)	**11 passed**	Real browser against the real backend: score, what-if studio, finance, bill upload, chat with verified numbers, mobile navigation without horizontal overflow.
API contract	**23 / 23**	Every field the UI reads exists in the live API response.
Static checks	**clean**	TypeScript strict, ESLint with zero warnings, Ruff.

Honest gaps. Four things were tested only against mocked responses because the build environment blocks those hosts: live Supabase sign-up, the live Nemotron call, the Telegram bot, and Gemini bill reading. A small Supabase step (`007_move_rls_helpers`) is still waiting to be run in the SQL editor. Data is synthetic plus public aggregates, so scores are estimates, and the interface says so.

4 Plan for the remaining time

When	Work	Done when
Next 2 hours	Run `connect_supabase` on the laptop; verify live sign-up and sign-in; run `llm_eval.py --llm` against Nemotron and record groundedness and latency; check the Telegram bot and a real PDF bill.	All four live paths pass, numbers added to the Model Lab.
Before demo	Apply migration 007; tune thinking budget for faster replies; rehearse the 5-screen story (bill → score → what-if → finance → Saathi) on a phone and laptop.	Median chat reply fast enough for a live demo; no console errors.
Stretch	Distil verified Nemotron answers into the SFT set (`--distill`) and fine-tune with NeMo Customizer; validate the model on a handful of real bills; add a lender view with real-data calibration bands.	Held-out groundedness at least as high as the template baseline with richer language.

Main risks: live reasoning latency (mitigation: thinking budget setting and template fallback); free-tier Supabase pausing after 7 days idle (restore from dashboard); synthetic data optimism (mitigation: real-data calibration views and honest confidence labels).